GLENCORE

Data Standard

Bill of Materials

12

Document properties

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1. Introduction

1.1 Master Data Standards - Purpose

The Data Standard provides clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 Bill of Materials Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for SAP Bill of Materials at Glencore.
- Ensure alignment with the Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Element 5
- Identify the minimum requirements for creating, maintaining, and retiring a Bill of Materials
- Provide the guiding principles to manage Bill of Materials Master Data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is targeted at the following audience:

- Master Data maintenance teams in order to understand how the object needs to be maintained
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to; Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard
- Transition to support team to assist in the development of user guides

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the governance for all requirements described in this Standard as well as approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders

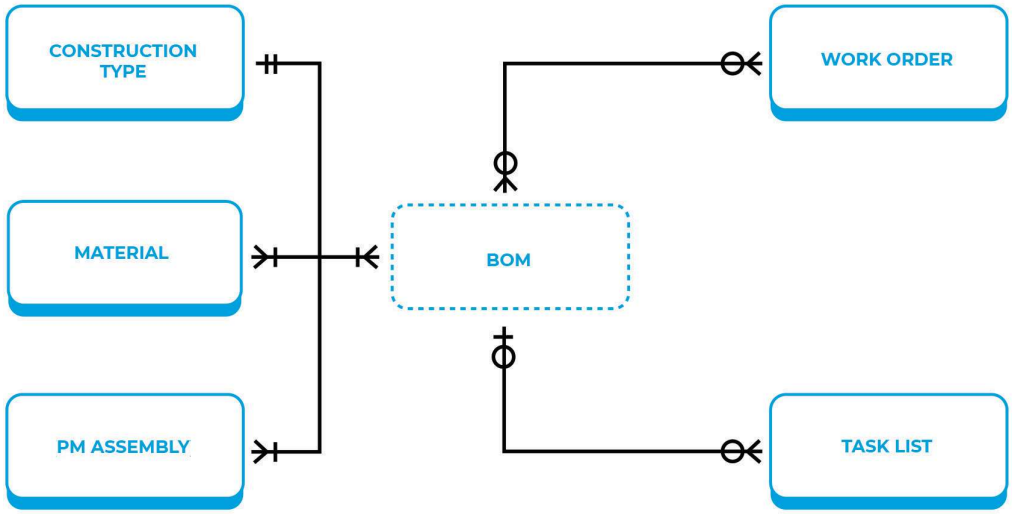
2. Data Object Description

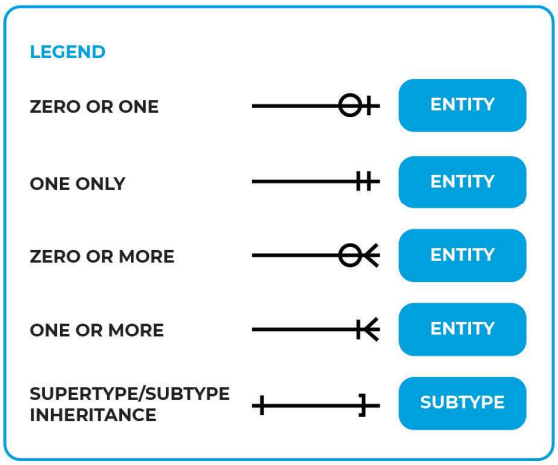
2.1 Definition

2.1.1 Maintenance Bill of Materials

A maintenance Bill of Materials (BOM) is a formally structured list of the components that make up a technical object or an assembly. The list contains the Material numbers of the individual components together with their quantity and unit of measure. The components can be catalogued stock or non-stock (Order on Demand) spares or assemblies, which in turn can be described using maintenance BOMs. See also Data Standard - Construction Type & PM Assembly.

2.2 Entity Relationship Diagram (ERD)





3. Business Requirements

3.1 Business Use

3.1.1 Maintenance BOMs

Maintenance BOMs are primarily used for the assignment of spare parts associated with a Functional Location or Equipment in a Work Order. The majority of BOMs will be created to support task lists identifying materials required for scheduled maintenance activities. For example:

- Materials required for a 500 hour service
- Materials required for an engine change out
- Materials required for pulley change out
- Materials for gearbox rebuilds

The secondary purpose of a BOM is to identify assembly information that allows materials to be recognised in a format similar to OEM parts manuals. BOMs of this type will be primarily used for reactive or corrective type tasks.

Components that are considered to be consumables are to be identified as part of the scheduled maintenance requirements and included in the BOM.

Functional Locations and Equipment	A Maintenance Assembly representing the structure of an item of plant is to be entered as a Construction Type in the Functional Location or Equipment master record. This linkage is displayed in the technical object structure display along with the material BOM that has been created for the Construction Type.
Work Orders	BOMs associated with the Functional Location, Equipment or Assembly in the Work Order header are available for a structured spare parts search in the Work Order.
Task Lists	A Maintenance Assembly entered in the header assembly of a Task List will provide a list of components for selection in the Task List Operation if it has also been defined as a Material BOM.

3.2 Business design Rules & Criteria

The following business and design rules are to be applied in determining how to structure a BOM:

- Only Material BOMs will be used. There will be no Functional Location or Equipment BOMs.
- All BOMs shall be associated with a Functional Location or Equipment via a Construction Type. There will be no BOMs that are not linked to a Construction Type.
- Maintenance BOMS will be usage type '4'; Plant Maintenance.

3.2.1 BOMs for different Plants

Material BOMs are Plant specific. However, Construction Types should be common for similar Physical Assets across the organisation. Therefore, there will be multiple Material BOMs for one Construction Type if the Physical Asset is used at more than one Plant.

The basic structure of the PM Assemblies (sub-assemblies) of a Construction Type BOM is to be consistent across Plants for the first level of the BOM. Material components may differ between sites (see example structure in Hierarchy Requirements).

Plant specific data may be entered in BOM Item Text 1 and 2. The information contained in these texts is to enable more specific searches where the Material Short Description does not completely describe the item.

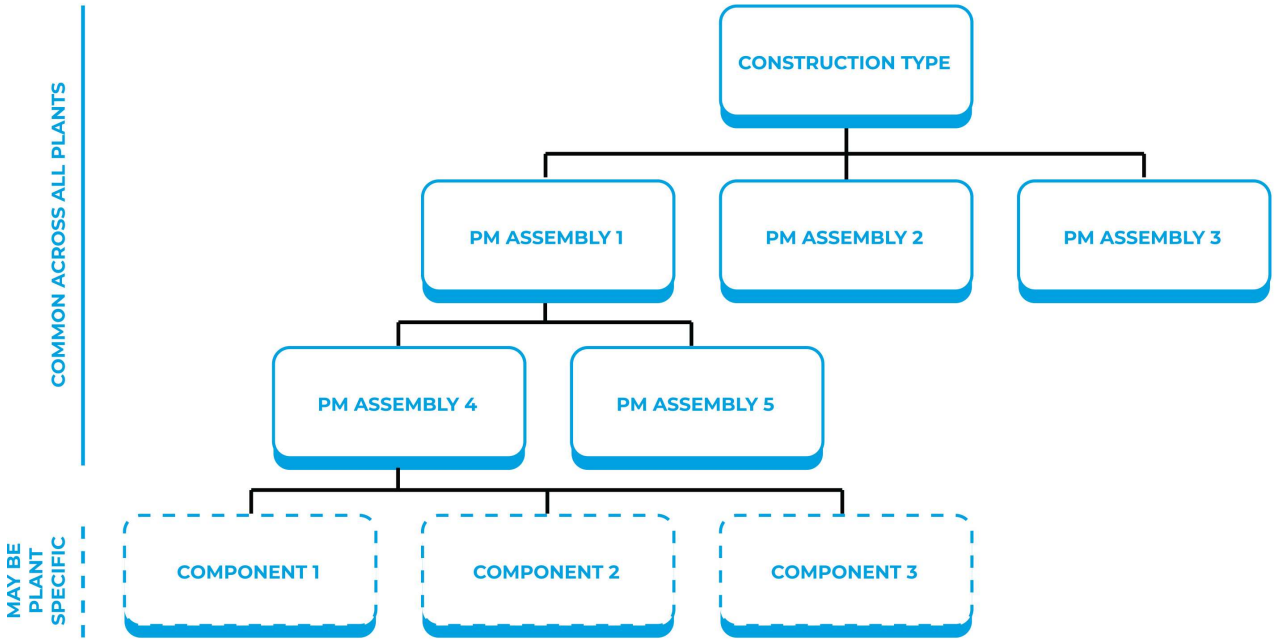
3.2.2 BOM Components

Quantity of issue in the BOM component should reflect the installed quantity. When the BOM component is assigned to a Task List, the quantity shall reflect the actual quantity required for the specific task.

Material BOMs shall not contain Non-Catalogued Materials. That is, every BOM item will have a material number.

3.3 Hierarchy Requirements

An example structure of a Material BOM is shown below:



Items in a Material BOM for a Construction Type will always be Maintenance Assemblies of type PM Assembly. No direct components are to be assigned at this level.

A PM Assembly can contain either further PM Assemblies or direct components, but not both.

The PM Assemblies should be grouped to represent logical groups of components.

There is no limit to the number of levels allowed. However, usability must be considered when determining the width (components or PM Assemblies per level) and depth of the BOM.

4. Detail Standard

4.1 Data Object Coding Standard

4.1.1 Internal numbering

A BOM is uniquely identified by a BOM Category, Bill of Material number and BOM Alternative. BOM Category will always be 'M' for Material BOM. Bill of Material number is an internally (SAP system) generated number (only relevant to the system) and the Alternative BOM for Maintenance BOMs is always '1'.

4.1.2 External numbering

Externally, a BOM is uniquely identified by Plant, Material number and BOM Usage. Material number shall be either a Maintenance Assembly of type Construction Type or PM Assembly, or another catalogued Material for which a BOM has been created. BOM Usage will be '4' for a Plant Maintenance BOM.

4.2 Attributes

4.2.1 BOM Header Usage

Application Definition	This key defines the area where a BOM can be used.
Application Guiding Principles	The BOM Usage separates the BOM according to the department in which they are used. Usage of Material BOMs can be shared.
Guiding Principles	BOMs shall be categorised only for Plant Maintenance usage.
Application Coding Convention	1 character alphanumeric
Naming Convention	Plant Maintenance – Usage '4'

4.2.2 BOM Item Category

Application Definition	Item Category
Application Guiding Principles	Categorisation of the items in a BOM according to set criteria, such as whether they refer to an object (e.g. material master) or whether they are kept in stock.
Guiding Principles	As per Application Guiding Principles.
Application Coding Convention	1 character alphanumeric

Naming Convention	Glencore will use the following item categories only: L – Catalogued material I – Maintenance Assembly (material type IBAU) N – Non-stock material (Not Available)

4.2.3 BOM Item number

Application Definition	This number refers to the sequence of BOM components.
Application Guiding Principles	The item number increment is defined in customising.
Guiding Principles	Default component numbering will be in increments of 0010. User-specific BOM item number increments shall not be used.
Application Coding Convention	4 character numeric
Naming Convention	Initial creation of BOM items will use the BOM increments of 0010. BOM Items added subsequently that are required to be inserted between existing numbers, shall use 0005, 0015 etc.

4.2.4 BOM Item Text 1 / 2

Application Definition	Descriptive Text.
Application Guiding Principles	Text of up to 40 characters, which forms the first of two lines of text describing the item.
Guiding Principles	Additional Plant Specific text that may be used to describe the item more fully or colloquially to aid in the selection of the correct component in a work order.
Application Coding Convention	40 character alphanumeric
Naming Convention	Refer to Data Standard – General Configuration

4.3 Hierarchy Standards

See Section Hierarchy Requirements.

5. Appendix A: Additional Information

5.1 Complete Field List

Legend for Attribute Table

- **Table –** SAP table where data object appears
- **Field –** SAP data attribute field
- **Field Name –** SAP field name of data object
- **Description –** Description of input required
- **Data Type –** SAP data type (CHAR=Character, DATS=Date, QUAN=Quantity, UNIT=Measurement type)
- **Length –** Length of field
- **MOID –** Data requirement (M=Mandatory, O=Optional, I=Internally populated, D=Data Origin i.e. Inherited from the Material Master)
- **Field Validation Type –** Entry type allowed (e.g. Free text, List selection) and additional information

5.1.1 BOM Header

Table	Field	Field Name	Description	Data Type	Len	M/O/I/D	Field Validation Type
RC29K	STLTY	BOM category	Categorises BOM according to object for which the BOM is created. For example, Equipment, Functional Location or Material.	CHAR	1	I	Always 'M' for Material BOM
RC29N	DATUV	Valid-From Date	Date from which BOM is valid.	DATS	8	M	BOMs manually created after the initial load shall use the default of today's date.
RC29K	WERKS	Plant	Key uniquely identifying a plant	CHAR	4	M	5030 – Cobar 7490 – Ernest Henry 8110 – Mount Isa Mine 8111 – Townsville Port 9830 – Copper Refinery
RC29K	STLNR	Bill of material	Together with the BOM Category and the Alternative BOM, identifies the Bill of Material.	CHAR	8	I	The system assigns this number internally. These numbers are irrelevant to the user.
RC29K	STLAL	Alternative BOM	Identifies one BOM in a BOM group.	CHAR	2	M	Always 1 for Maintenance BOMs
RC29K	STLAN	BOM Usage	This key defines the area where a BOM can be used	CHAR	1	M	Always 4 – Plant Maintenance

Table	Field	Field Name	Description	Data Type	Len	M/O/I/D	Field Validation Type
RC29K	BMENG	Base quantity	Quantity to which all the BOM components relate.	QUAN	13	M	Default quantity of 1
RC29K	BMEIN	Base unit of measure for BOM	Used as the base unit for all component quantities in this BOM	UNIT	3	M	Default unit of EA

5.1.2 BOM Item

Table	Field	Field Name	Description	Data Type	Len	M/O/I/D	Field Validation Type
RC29P	POSNR	BOM Item Number	This number refers to the sequence of BOM components	CHAR	4	M	Default component numbering will be in increments of 0010. BOM Items added subsequently that are required to be inserted between existing numbers, shall use 0005, 0015 etc
RC29P	POSTP	Item Category (Bill of Material)	Categorization of the items in a BOM according to set criteria, such as whether they refer to an object (for example, material master) or whether they are kept in stock.	CHAR	1	M	Glencore will use the following item categories: L – Catalogued material I – Maintenance Assembly (material type IBAU)
RC29P	IDNRK	BOM component	Material number for BOM items	CHAR	7	M	Table Selection.

Table	Field	Field Name	Description	Data Type	Len	M/O/I/D	Field Validation Type
RC29P	MENGE	Component quantity	Quantity of the component, which is entered in relation to the installed quantity	QUAN	13	M	Quantity shall be the installed quantity, or '1' if this is not known.
RC29P	MEINS	Component unit of measure	The unit for the component quantity	UNIT	3	D	The Unit of Measure has been specified in the Material Master record and shall default here.
RC29P	POTX1	BOM Item Text (Line 1)	Text of up to 40 characters, which forms the first of two lines of text describing the item.	CHAR	40	O	Additional Short text description for each BOM Item as required.
RC29P	POTX2	BOM item text (line 2)	Text of up to 40 characters, which forms the second of two lines of text describing the item.	CHAR	40	O	Additional Short text description for each BOM Item as required.

6. Stakeholder Signoff

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